GREENOPS DIGITAL INFRASTRUCTURE AUDIT PACK

Tenant Identity: enterprise-bank-uk | **Generated:** 2026-10-22 21:51:16 UTC | **Assurance Framework:** ISAE 3410 / ISO 14064-2 | **Ledger Security:** Ed25519 PKI Sealed

1. Executive Energy & Emissions Summary

Metric	Value	Accounting Standard / Method
Total Workloads Processed	2	Batch Compute Executions
Total Energy Consumed	22.62 kWh	Direct Hardware & Cluster Metering
Statutory Scope 2 Emissions	3.14 kg CO2e	UK SECR (NESO AEF Location-based)
Consequential Carbon Displaced	0.69 kg CO2e	Elexon BMRS Marginal Emissions Factor (MEF)
Estimated Spot Cost Savings	£0.45	Wholesale Grid / Off-Peak Spot Shift Arbitrage
SLA Continuity Overrides	0 (0.0%)	Zero-Downtime Pipeline Safety Invariant Active

2. Non-Repudiable Settlement Ledger Blocks (Ed25519 Signed)

Settlement Window (UTC)	Jobs	kWh	Merkle Root Hash	Ed25519 PKI Signature
2026-10-22 22:51:16	2	100.00	abbf72d72d60329b...	1jsfJdSY5Gh4qvyT...
2026-10-22 22:21:12	1	150.00	0b83afd89ea8996c...	+H8UlHOHmmgQQsrK...

3. Statutory Methodology & Auditor Notes

Statutory SECR Scope 2 Compliance: Calculated using location-based Average Emissions Factors (AEF) published by NESO. Compliant with UK Companies Act 2006 Regulations 2018.
Cryptographic Provenance: All settlement blocks are aggregated into deterministic binary Merkle trees and signed with an Ed25519 private key. Auditor verification public keys are distributed out-of-band.